Supplementary tables

Full audited tables. Main-text percentages do not use the Expt-9 and Expt-10 primitive rows whose recorded saving is zero. Both Expt-8C nets stay in this supplement. Table 8C separates the mean of per-box retention ratios from the ratio of summed savings.

Table 1. Experimental design and scope of inference across memory-reuse mechanisms

Mechanism	Experiments	Source of memory	Workload	Constructed intervention	Learning / evaluation separation	Principal question
Primitive packing and early stored memory	Expt-1, Expt-2, Expt-2A, Expt-3	Empty, short-term, or long-term stores; Expt-2 discovers flat operations during search	Repeated 10×10 inventories; Expt-2 held-out inventories	None in Expt-1. Expt-2 search is autonomous within a typed action language	Expt-2 library is frozen before its held-out comparison (Expt-2A)	Does stored or discovered memory change occupied cells once deliberation is counted?
Learned flat and hierarchical operations	Expt-4 through Expt-8C	Mined operations, selective libraries, consult/commit gates, an index, and action-level predictors	Held-out packing boxes; Expt-8C uses 50 new boxes	None at deployment. Expt-6 and Expt-8C use diagnostic remaining-solves	Libraries and gates are frozen before the reported evaluation. Expt-8B's eight-box holdout is development data, not Expt-8C	Can a learned operation be selected so that whole-box effort falls without losing quality?
Exact continuation recall	Expt-9, Expt-10	One shared first-write table of exact remaining states	Stored inventories, fresh inventories, and, in Expt-10, labeled intermediate prefixes	Expt-10 intermediate prefixes are supplied and the covering is committed	The table is frozen before fresh evaluation. First write wins; seeds are not separate memories	Does exact recall replace search when the same state recurs?
Supplied and experience-selected transformations	Expt-11, Expt-12	The same first-write table, plus a declared set of rotations and reflections	Constructed transformed remainders; Expt-12 mixes them into hidden streams at predeclared rates	Transformed suffixes and slot categories are supplied. The category is hidden at test in Expt-12	Which transforms are kept is frozen before evaluation. Expt-12's selected set is not the Expt-11 set	Can experience choose among supplied equivalences, and does that help when recurrence is partial?
Unforced recurrence and online growth	Expt-13, Expt-14	Frozen exact table (Expt-13 also keeps the selected transforms). Expt-14 may insert its own solved states	Ordinary fresh and family streams, with no scheduled matches	None on the evaluation streams. A coverage pass is used only to select transforms before Expt-13 evaluation	Selection is frozen before Expt-13. Expt-14 grows one table per stream after each solve	Do ordinary trajectories meet a stored intermediate state, and does writing new states pay for itself?
Local-region acquisition and reuse	Expt-15, Expt-16	Exact fillings of bounded free regions	Supplied regions in Expt-15 Test 1; ordinary trajectories in Test 2 and Expt-16	Test 1 assigns regions and budgets. Expt-16 searches recorded late regions offline	Entries are frozen before evaluation. Expt-16 counterfactuals are diagnostic and below the recurrence threshold	Are late free regions a usable source of exact local solutions?
Recurring arrangements, diagnostic utility, and	Expt-17, Expt-18, Expt-19	Connected groups of two or three pieces from acceptable primitive	Fresh and family manifests. Expt-19's decision set is validation	None. Expt-18 replays sampled legal placements	The dictionary is frozen before fresh examination. Expt-19's budget is set on	Do small successful arrangements recur, are they useful if selected with

Mechanism	Experiments	Source of memory	Workload	Constructed intervention	Learning / evaluation separation	Principal question
bounded recognition		packings; translation removed by a declared rule	fresh	as a diagnostic oracle	development manifests before validation	hindsight, and can this recognizer expose them within the cost gate?

Notes.

Table 2A. Expt-4C held-out quality and effort for selective libraries

Experiment	Condition	Sample structure	Mean Q	Mean A	Paired saving	Saving (%)	Quality-loss rate
Expt-4C	Primitive baseline	100 tasks × 10 seeds	75.45	1,611.6	0.0	0.0%	0.0%
Expt-4C	Selective flat library	100 tasks × 10 seeds (10 libraries)	75.42	1,923.1	-311.6	-19.3%	11.0%
Expt-4C	Selective hierarchical library	100 tasks × 10 seeds (10 libraries)	75.14	2,250.7	-639.1	-39.7%	14.5%

Notes.

Table 2B. Expt-5 held-out quality and effort for consult and commit gates

Experiment	Condition	Sample structure	Mean Q	Mean A	Paired saving	Saving (%)	Quality-loss rate
Expt-5	Primitive baseline	16 eval tasks × 3 seeds (3 libraries)	75.60	1,442.2	0.0	0.0%	0.0%
Expt-5	Current selective library	16 eval tasks × 3 seeds (3 libraries)	75.40	2,157.8	-715.5	-49.6%	4.2%
Expt-5	Consult gate only	16 eval tasks × 3 seeds (3 libraries)	75.60	1,821.0	-378.8	-26.3%	0.0%
Expt-5	Commit gate only	16 eval tasks × 3 seeds (3 libraries)	75.44	2,170.4	-728.1	-50.5%	4.2%
Expt-5	Both gates	16 eval tasks × 3 seeds (3 libraries)	75.60	1,821.0	-378.8	-26.3%	0.0%

Notes.

Table 2C. Expt-7 held-out quality and effort for indexed retrieval

Experiment	Condition	Sample structure	Mean Q	Mean A	Paired saving	Saving (%)	Quality-loss rate

Experiment	Condition	Sample structure	Mean Q	Mean A	Paired saving	Saving (%)	Quality-loss rate
Expt-7	Primitive baseline	48 paired eval boxes	75.54	1,449.1	0.0	0.0%	0.0%
Expt-7	Frozen gates, no index	48 paired eval boxes	75.54	1,838.1	-389.0	-26.8%	0.0%
Expt-7	Indexed retrieval	48 paired eval boxes	75.27	1,656.1	-207.0	-14.3%	10.4%
Expt-7	Indexed ranker	48 paired eval boxes	75.54	1,471.3	-22.2	-1.5%	0.0%

Notes. Each Expt-7 seed uses its own frozen library. Snapshot hashes for seeds 11, 22, and 33 differ, and the run copies match those snapshots.

Table 3A. Expt-8C frozen predictors on unseen action-level cases

Predictor	Evaluated actions	MAE	Accepted actions	Accepted unique boxes	Contributing seeds	Mean remaining saving	Mean net after site checks	Quality-loss rate	Severe-loss rate
per_seed_mean	541	1,152.6	231	40	33	203.8	201.5	10.8%	26.8%
frozen (historical)	541	1,357.5	0	0	Not applicable	Not recorded	Not recorded	Not recorded	Not recorded
context_unweighted	541	1,151.4	4	3	33	-415.0	-418.2	25.0%	25.0%
action_unweighted	541	1,188.5	8	7	33	24.5	21.2	0.0%	12.5%
context_weighted (historical)	541	2,688.3	0	0	Not applicable	Not recorded	Not recorded	Not recorded	Not recorded
action_weighted (historical)	541	2,848.9	0	0	Not applicable	Not recorded	Not recorded	Not recorded	Not recorded

Notes.

Table 3B. Expt-8C action-model cost ledger, including rejected candidates

Component	Value	What is counted
Gross accepted remaining saving	196.0	Sum of remaining-solve savings on the 8 accepted actions
Accepted-action decision cost in the saved ledger	209.0	Site checks and feature checks on accepted actions, as stored in offline.json
Consultation cost on every scored state	1,608.0	Index and ranking, including states whose action was rejected
Feature checks in the cost ledger	17,167	All first-legal action_vector cell inspections, accepted and rejected
Saved offline net (accepted features only)	-1,621.0	Ledger field. The report states this subtracts accepted features only

Component	Value	What is counted
Decision-relevant net including rejected features	-19,893	Report-derived: accepted saving minus accepted site checks, all feature checks, rejected site checks, and all consultation

Notes.

Table 4A. Expt-9 exact continuation recall

Experiment and task	Condition	Paired evaluations	Unique tasks	Mean A	Comparator A	Effort saving	Mean Q	Quality-loss rate	Initial hits	Intermediate hits	Validation failures
Recurring stored inventories	Exact memory	36	12	16.8	885.8	868.9	76.75	0.0%	Not recorded	Not recorded	0
Store primitive solves	Primitive	36	12	900.8	885.8	0.0	76.36	0.0%	Not recorded	Not recorded	0
Fresh inventories	Exact memory	36	12	1,267.1	1,250.8	-16.4	75.97	0.0%	Not recorded	Not recorded	0
Fresh inventories	Primitive	36	12	1,250.8	1,250.8	0.0	75.97	0.0%	Not recorded	Not recorded	0

Notes.

Table 4B. Expt-10 exact subproblem recall, including labeled intermediate prefixes

Experiment and task	Condition	Paired evaluations	Unique tasks	Mean A	Comparator A	Effort saving	Mean Q	Quality-loss rate	Initial hits	Intermediate hits	Validation failures
Recurring stored inventories	Exact memory	36	12	16.5	1,238.0	1,221.5	75.92	5.6%	36	Not recorded	0
Constructed nonempty prefixes	Memory on	30	10	181.3	1,611.5	1,430.2	76.00	0.0%	0	Not recorded	0
Constructed nonempty prefixes	Memory off	30	10	1,611.5	1,611.5	0.0	75.30	0.0%	0	Not recorded	0
Fresh inventories	Exact memory	36	12	1,680.0	1,663.1	-16.9	75.56	0.0%	0	Not recorded	0

Experiment and task	Condition	Paired evaluations	Unique tasks	Mean A	Comparator A	Effort saving	Mean Q	Quality-loss rate	Initial hits	Intermediate hits	Validation failures
Fresh inventories	Primitive	36	12	1,663.1	1,663.1	0.0	75.56	0.0%	0	Not recorded	0

Notes.

Table 4C. Expt-11 supplied transformations and experience-selected transforms

Experiment and task	Condition	Paired evaluations	Unique tasks	Mean A	Comparator A	Effort saving	Mean Q	Quality-loss rate	Initial hits	Intermediate hits	Validation failures
Transformed test remainders	Exact match	33	11	1,003.1	990.8	-12.3	75.52	0.0%	0	Not recorded	0
Transformed test remainders	All supplied transforms	33	11	184.3	990.8	806.5	76.00	0.0%	0	Not recorded	0
Transformed test remainders	Learned selection	33	11	184.3	990.8	806.5	76.00	0.0%	0	Not recorded	0
Transformed test remainders	Primitive	33	11	990.8	990.8	0.0	75.52	0.0%	0	Not recorded	0
Fresh inventories	Learned selection	36	12	1,886.0	1,835.3	-50.7	75.50	0.0%	0	Not recorded	0
Fresh inventories	Primitive	36	12	1,835.3	1,835.3	0.0	75.50	0.0%	0	Not recorded	0

Notes.

Table 5. Expt-12 net effort in constructed mixed streams

Predeclared recurrence	Realized constructed share	Observed reuse	Primitive mean A	Exact-memory saving	Supplied-transform saving	Learned-selection saving	Quality-loss rate
0.0	0.0%	1.4%	1,399.6	-14.7	-95.5	-37.5	0.0%
0.05	4.2%	5.6%	1,981.5	-15.5	-47.2	11.9	0.0%
0.1	8.3%	8.3%	1,466.4	45.5	-8.5	47.0	0.0%

Predeclared recurrence	Realized constructed share	Observed reuse	Primitive mean A	Exact-memory saving	Supplied-transform saving	Learned-selection saving	Quality-loss rate
0.25	25.0%	25.0%	1,553.7	219.2	309.1	354.4	0.0%
0.5	50.0%	51.4%	1,810.8	502.0	857.2	887.1	0.0%
1.0	100.0%	100.0%	1,404.2	659.8	1,319.2	1,319.2	0.0%

Notes.

Table 6A. Expt-13 unforced reuse on ordinary streams

Workload	Memory condition	Streams	Boxes per stream	Initial hits	Intermediate hits	Mean cumulative saving per stream	Range across streams	Saving on never-encountered inventories	Quality losses
Independent fresh	Exact memory	3	24	0	0	-391.0	-395.0 to -387.0	-16.3	0
Independent fresh	Learned transforms	3	24	0	0	-1,564.0	-1,580.0 to -1,548.0	-65.2	0
Recurring families	Exact memory	3	36	12	0	4,727.0	3,545.0 to 6,932.0	-16.4	0
Recurring families	Learned transforms	3	36	12	0	3,464.7	2,069.0 to 5,525.0	-55.8	0

Notes.

Table 6B. Expt-14 chronological exact-memory growth

Workload	Memory condition	Streams	Boxes per stream	Initial hits	Intermediate hits	Mean cumulative saving per stream	Range across streams	Saving on never-encountered inventories	Initial keys	Final keys	Quality losses
Independent fresh	Frozen exact	6	30	0	0	-490.5	-504.0 to -481.0	-16.4	684	684	0
Independent fresh	Growing exact	6	30	0	0	-1,358.5	-1,397.0 to -1,312.0	-45.3	684	1089 to 1141	0
Recurring families	Frozen exact	6	30	11	0	1,129.3	-504.0 to 2,242.0	-16.3	684	684	0

Workload	Memory condition	Streams	Boxes per stream	Initial hits	Intermediate hits	Mean cumulative saving per stream	Range across streams	Saving on never-encountered inventories	Initial keys	Final keys	Quality losses
Recurring families	Growing exact	6	30	15	0	936.3	66.0 to 1,404.0	-45.9	684	1077 to 1109	0

Notes.

Table 7A. Expt-15 supplied-region component versus decomposed search and versus whole-box search

Comparison	Paired boxes	Mean Q of local policy	Mean A of local policy	Paired saving	Quality-loss rate
Local decomposed versus primitive decomposed, matched regions	72	75.65	1,207.2	534.0	0.0%
Local decomposed versus whole-box primitive, matched regions	72	75.65	1,207.2	15.2	11.1%

Notes.

Table 7B. Expt-16 acquisition of late-region fillings and read-only applicability

Quantity	Value	Counting rule
Regions examined	128	Recorded late regions in acceptable store trajectories
Exact fillings	5	Offline search found a complete filling
Area-impossible	52	Region size is not a sum of remaining piece sizes
Exhaustively impossible	71	Impossible count minus area-impossible count
Budget timeouts	0	Search stopped at the predeclared budget
Detection cost	20,684	Flood-fill cell visits during acquisition
Search cost	58,413	Candidate placements examined offline
Validation cost	13	Checks of found fillings
Insertion cost	10	Writes into the local table
Geometry matches, fresh	3	Read-only lookup on 60 fresh boxes

Quantity	Value	Counting rule
Inventory-compatible matches, fresh	1	Geometry match with the multiset available
Geometry matches, family	2	Read-only lookup on 60 family boxes
Inventory-compatible matches, family	1	Geometry match with the multiset available
Mean detector cost, fresh	497.8	Not primitive effort
Mean primitive effort, fresh	1,597.7	Same boxes, primitive policy
Mean detector cost, family	497.1	Not primitive effort
Mean primitive effort, family	1,138.2	Same boxes, primitive policy

Notes.

Table 7C. Expt-16 diagnostic counterfactuals at the two inventory-compatible matches

Workload	Task	Region size	Primitive remaining effort	Commit remaining effort	Saving	Quality change	Quality loss
independent_fresh	inv_09f4d3398327	8	192.0	98.0	94.0	0.00	no
recurring_family	inv_3c58e971670e	8	384.0	962.0	-578.0	-1.00	no

Notes.

Table 7D. Expt-15 ordinary streams, without supplied regions

Workload	Condition	Boxes	Mean Q	Mean A	Paired saving versus primitive	Quality-loss rate	Local hits
Independent fresh	Primitive	60	75.20	1,822.2	0.0	0.0%	0
Independent fresh	Exact whole-box memory	60	75.20	1,839.4	-17.1	0.0%	0
Independent fresh	Local-region memory	60	75.20	2,316.6	-494.3	0.0%	0
Recurring families	Primitive	60	75.98	1,118.3	0.0	0.0%	0
Recurring families	Exact whole-box memory	60	75.98	1,134.2	-15.9	0.0%	0
Recurring families	Local-region memory	60	75.98	1,623.3	-505.0	0.0%	0

Notes.

Table 8A. Expt-17 recurrence and legal-fit census

Quantity	Fresh	Family
Boxes	60	60
Boxes with any legal fit	59	60
Boxes with an early or middle fit	59	60
Early fit events	4,999	6,888
Middle fit events	2,089	2,206
Late fit events	19	36
Mean primitive effort	1,597.7	1,138.2
Mean matching checks	72,872.2	93,095.8
Ratio of mean matching checks to mean primitive effort	45.6	81.8

Notes.

Table 8B. Expt-18 diagnostic utility of a sampled set of legal placements

Quantity	Value
Sampled fresh boxes	12
Replayed fresh placements	91
Valid replays with quality loss beyond one cell	25
Valid replays	91
Boxes whose best quality-preserving saving meets the floor of 48	10
Of those, empty-board early	5
Of those, middle	5
Diagnostic discovery checks	5,764
Diagnostic counterfactual effort, both branches	342,742

Notes.

Table 8C. Expt-19 held-out recognition cost and sampled-reference recall

Quantity	Value
Fresh boxes	60
Mean recognition cost	127.7
Mean primitive effort	1,125.0
Ratio of means	0.113
Mean of per-box ratios	0.138
Predeclared cost gate	0.10 of primitive effort
State recall	0.623
Mean of per-box retention ratios	0.792
Useful actions exposed	69
Budget exhausted before the state	49
Budget exhausted during the consultation	24
Anchor misses among useful reference actions	0
Unlabeled exposures	198
Ratio of summed retained saving to summed oracle saving	0.744

Notes. The mean of per-box retention ratios is the mean, over 39 boxes with a useful reference action, of retained saving divided by that box's oracle saving. The ratio of summed savings is 12345.0 / 16589.0. These summaries are different.

Table S1. Run provenance, corrected results, and dataset reuse

Experiment	Authoritative directory	Config SHA-256	Role
Expt-1	Expt_01_10x10_Packing/results	Not recorded	environment smoke; read-only for later experiments
Expt-2 baseline	Expt_02_Autonomous_Abstraction/results/20260929_092518_discovery_smoke	19944168bb0212e392fb31af8c8b15873cac2c556c5805a79b6c39f4ef6e2cdd	authoritative discovery smoke; later cap-progress runs are not this baseline

Experiment	Authoritative directory	Config SHA-256	Role
Expt-2A	Expt_02_Autonomous_Abstraction/results/20260929_110932_expt2a	dd22d9c981c70357a3534da2a9b09aec27a631d390a76b3f22bdef366c308219	frozen-policy comparison
Expt-4C	Expt_04C_Confirm_Reusable_Knowledge/results/20260929_200422_confirm	f3d8eb2ae6bfa0f58e393575096a9e53aae9cecf5f8a72c6a64acf977b34db01	authoritative confirm; smoke 20260929_200405 superseded
Expt-5	Expt_05_Consult_and_Commit_Gates/results/20260929_212658_diagnostic	89502b2fb268751f519b089e7c8e5cd6bad306012e6e486b62d39731750c36db	authoritative; smoke superseded
Expt-7	Expt_07_Indexed_Retrieval/results/20260929_220030_diagnostic	20c07c79fc706adf8b34fabf778071e419f62c84ad243224eb6c4fd5c5d7d577	authoritative; 20260929_220009 lacks the frozen-gates cell
Expt-7B	Expt_07B_Decision_Audit/results/20260929_223742_diagnostic	e5c7fe56d7e97c24b4d857106ac77b64fb0baaf9087d6667a3644c6fb26b4978	audit of the frozen ranker
Expt-8C	Expt_08C_Unseen_Confirmation/results/20260929_233422_diagnostic	9a5d688da346b002920d2fea5f79b5e29e295b27b930a72902ba34a43d3e7ea3	unseen action-level confirmation
Expt-9	Expt_09_Exact_Continuation_Reuse/results/20260929_234344_diagnostic	69daa5c3977167b2b050ba4fd7776c7463b94a5e4f394cf0227b0eafb841847e	authoritative diagnostic
Expt-10	Expt_10_Exact_Subproblem_Reuse/results/20260930_055535_diagnostic	1c48099dffe05894159b8e352e5a0473386eac94182a37f8cfa22d9b959a675f	authoritative diagnostic
Expt-11	Expt_11_Remaining_State_Equivalence/results/20260930_060957_diagnostic	45ffae7a1bfa3b0fccf19faf23ce707ca3fea4f581b246bcc658079a160afbd5	authoritative diagnostic
Expt-12	Expt_12_Mixed_Stream_Recurrence/results/20260930_061916_diagnostic	a5da6de24d10bd1193f50d99fc26afd1839823adfef65020f7816dd8fe9fee58	authoritative diagnostic
Expt-13	Expt_13_Unforced_Reuse/results/20260930_064034_diagnostic	7efe29f8869c79397c5846462365bc4a5d1d34505279304aa89be07096237b3e	authoritative diagnostic
Expt-14	Expt_14_Online_Memory_Growth/results/20260930_064836_diagnostic	a5de033c8d484c06b9ccce19870446a2bc1cc5376e2c0b1e0b1e1605d43e0501	authoritative diagnostic
Expt-15	Expt_15_Local_Subproblem_Memory/results/20260930_070329_diagnostic	06fee83d54c3e57ee044edf53529d4359683a302b10a23b5ac5ce0f5fd9bb584	authoritative; 20260930_065912 differs on fresh streams
Expt-16	Expt_16_Search_Acquired_Local_Memory/results/20260930_071552_diagnostic	fa332f6d9783881d858296b4f073f7fecdd2adf11b29cc3a17b5267f0e7ba927	corrected decision row;

Experiment	Authoritative directory	Config SHA-256	Role
			20260930_071519 superseded only for that row
Expt-17	Expt_17_Arrangement_Census/results/20260930_090000_diagnostic	b0c0587b4364400f20c21633d20316d26468a33e0f744fceee48db9c3f614415	authoritative census
Expt-18	Expt_18_Arrangement_Utility/results/20260930_091301_diagnostic	a9f727d5d03ea7c53e6a2556a40278584da4b0378d4884d2d0054926d300c0b1	authoritative utility sample; smokes are pipeline checks
Expt-19	Expt_19_Budgeted_Recognition/results/20260930_092450_diagnostic	9461d8e80cc01f62dd7e0363f894a815531c713c1d05ed53ecad97cb4e298d1a	authoritative held-out recognition; smoke superseded
Expt-17 frozen dictionary	Expt_17_Arrangement_Census/results/20260930_090000_diagnostic/frozen.json	b02cd72c92b7a9f00b8660b7a54633a1cb93844c62d20ff98ae0812ea62c904c	Frozen arrangement dictionary reused by Expt-18 and Expt-19. This hash is the file hash, not a second experiment.
Shared manifests	Expt_16_Search_Acquired_Local_Memory/manifests		Expt-16 through Expt-19 reuse these frozen task lists. Validation fresh does not overlap applicability fresh. Eight validation-family ids also occur in applicability family and were excluded from the Expt-19 family score.

Notes.

Table S2A. Expt-4C per-seed paired results

Condition	Seed	Boxes	Wins	Quality losses	Mean paired ΔA
selective_flat	11	100	13	9	-302.2
selective_flat	22	100	19	12	-387.5
selective_flat	33	100	12	13	-508.0

Condition	Seed	Boxes	Wins	Quality losses	Mean paired ΔA
selective_flat	44	100	19	9	-169.6
selective_flat	55	100	18	12	-347.3
selective_flat	66	100	18	11	-265.4
selective_flat	77	100	28	14	-318.9
selective_flat	88	100	20	7	-177.8
selective_flat	99	100	20	13	-166.7
selective_flat	111	100	9	10	-472.2
selective_hierarchical	11	100	12	6	-306.0
selective_hierarchical	22	100	16	15	-786.0
selective_hierarchical	33	100	15	16	-658.5
selective_hierarchical	44	100	19	15	-792.9
selective_hierarchical	55	100	14	15	-857.2
selective_hierarchical	66	100	21	17	-509.2
selective_hierarchical	77	100	22	20	-977.0
selective_hierarchical	88	100	16	15	-369.1
selective_hierarchical	99	100	6	12	-751.4
selective_hierarchical	111	100	20	14	-384.0

Notes.

Table S2B. Expt-5 per-seed paired evaluation

Experiment	Condition	Seed	Boxes	Mean Q	Mean A	Paired saving	Quality-loss rate
Expt-5	both_gates	11	16	75.56	1,844.7	-252.7	0.0%
Expt-5	both_gates	22	16	75.69	1,945.5	-485.1	0.0%
Expt-5	both_gates	33	16	75.56	1,672.9	-398.7	0.0%
Expt-5	current_selective	11	16	75.56	2,055.9	-463.9	0.0%

Experiment	Condition	Seed	Boxes	Mean Q	Mean A	Paired saving	Quality-loss rate
Expt-5	current_selective	22	16	75.69	1,929.1	-468.6	0.0%
Expt-5	current_selective	33	16	74.94	2,488.2	-1,214.0	12.5%
Expt-5	primitive	11	16	75.56	1,592.0	0.0	0.0%
Expt-5	primitive	22	16	75.69	1,460.4	0.0	0.0%
Expt-5	primitive	33	16	75.56	1,274.2	0.0	0.0%

Notes.

Table S2C. Expt-13 per-stream cumulative paired savings

Workload	Condition	Stream	Boxes	Cumulative saving	Intermediate hits
independent_fresh	exact	0	24	-391.0	0
independent_fresh	exact	1	24	-395.0	0
independent_fresh	exact	2	24	-387.0	0
independent_fresh	learned	0	24	-1,564.0	0
independent_fresh	learned	1	24	-1,580.0	0
independent_fresh	learned	2	24	-1,548.0	0
recurring_family	exact	0	36	3,545.0	0
recurring_family	exact	1	36	6,932.0	0
recurring_family	exact	2	36	3,704.0	0
recurring_family	learned	0	36	2,800.0	0
recurring_family	learned	1	36	5,525.0	0
recurring_family	learned	2	36	2,069.0	0

Notes.

Table S2D. Expt-7 per-condition seed means

Split	Condition	Seed	Boxes	Mean Q	Mean A	Paired saving	Quality-loss rate

Split	Condition	Seed	Boxes	Mean Q	Mean A	Paired saving	Quality-loss rate
	both_gates_frozen	11	16	75.06	2,161.9	-309.1	0.0%
	both_gates_frozen	22	16	75.62	2,043.1	-465.6	0.0%
	both_gates_frozen	33	16	75.94	1,309.2	-392.2	0.0%
	indexed_ranked	11	16	75.06	1,876.9	-24.1	0.0%
	indexed_ranked	22	16	75.62	1,596.7	-19.2	0.0%
	indexed_ranked	33	16	75.94	940.3	-23.3	0.0%
	indexed_retrieval	11	16	74.31	2,511.2	-658.4	18.8%
	indexed_retrieval	22	16	75.88	1,053.1	524.4	0.0%
	indexed_retrieval	33	16	75.62	1,404.1	-487.1	12.5%
	primitive	11	16	75.06	1,852.8	0.0	0.0%
	primitive	22	16	75.62	1,577.5	0.0	0.0%
	primitive	33	16	75.94	917.0	0.0	0.0%

Notes.

Table S2E. Expt-9 per-seed paired evaluation

Split	Condition	Seed	Boxes	Mean Q	Mean A	Paired saving	Quality-loss rate
fresh	exact	11	12	76.17	1,263.6	-15.9	0.0%
fresh	exact	22	12	75.75	1,084.1	-16.5	0.0%
fresh	exact	33	12	76.00	1,453.7	-16.7	0.0%
fresh	primitive	11	12	76.17	1,247.7	0.0	0.0%
fresh	primitive	22	12	75.75	1,067.6	0.0	0.0%
fresh	primitive	33	12	76.00	1,437.0	0.0	0.0%

Split	Condition	Seed	Boxes	Mean Q	Mean A	Paired saving	Quality-loss rate
recurring	exact	11	12	76.75	16.8	884.8	0.0%
recurring	exact	22	12	76.75	16.8	958.8	0.0%
recurring	exact	33	12	76.75	16.8	763.2	0.0%
store	primitive	11	12	76.75	917.5	0.0	0.0%
store	primitive	22	12	76.50	990.4	0.0	0.0%
store	primitive	33	12	75.83	794.6	0.0	0.0%

Notes.

Table S2F. Expt-10 per-seed paired evaluation

Split	Condition	Seed	Boxes	Mean Q	Mean A	Paired saving	Quality-loss rate
constructed	memory_off	11	10	75.40	1,580.7	0.0	0.0%
constructed	memory_off	22	10	75.10	1,576.1	0.0	0.0%
constructed	memory_off	33	10	75.40	1,677.6	0.0	0.0%
constructed	memory_on	11	10	76.00	181.3	1,399.4	0.0%
constructed	memory_on	22	10	76.00	181.3	1,394.8	0.0%
constructed	memory_on	33	10	76.00	181.3	1,496.3	0.0%
fresh	exact	11	12	75.75	1,637.6	-16.5	0.0%
fresh	exact	22	12	75.75	1,621.5	-17.2	0.0%
fresh	exact	33	12	75.17	1,780.9	-17.0	0.0%
fresh	primitive	11	12	75.75	1,621.1	0.0	0.0%
fresh	primitive	22	12	75.75	1,604.2	0.0	0.0%
fresh	primitive	33	12	75.17	1,763.9	0.0	0.0%
recurring	exact	11	12	75.92	16.5	970.2	0.0%
recurring	exact	22	12	75.92	16.5	1,038.8	8.3%
recurring	exact	33	12	75.92	16.5	1,655.6	8.3%

Split	Condition	Seed	Boxes	Mean Q	Mean A	Paired saving	Quality-loss rate
store	primitive	11	12	75.92	1,002.2	0.0	0.0%
store	primitive	22	12	75.83	1,069.9	0.0	0.0%
store	primitive	33	12	75.08	1,684.2	0.0	0.0%

Notes.

Table S2G. Expt-14 per-stream cumulative paired savings

Workload	Condition	Stream	Seed	Boxes	Mean saving per box	Cumulative saving	Quality losses
independent_fresh	frozen_exact	0	11	30	-16.3	-489.0	0
independent_fresh	frozen_exact	1	22	30	-16.1	-484.0	0
independent_fresh	frozen_exact	2	33	30	-16.7	-502.0	0
independent_fresh	frozen_exact	3	44	30	-16.8	-504.0	0
independent_fresh	frozen_exact	4	55	30	-16.1	-483.0	0
independent_fresh	frozen_exact	5	66	30	-16.0	-481.0	0
independent_fresh	growing_exact	0	11	30	-45.6	-1,367.0	0
independent_fresh	growing_exact	1	22	30	-45.3	-1,360.0	0
independent_fresh	growing_exact	2	33	30	-43.7	-1,312.0	0
independent_fresh	growing_exact	3	44	30	-45.3	-1,358.0	0
independent_fresh	growing_exact	4	55	30	-46.6	-1,397.0	0
independent_fresh	growing_exact	5	66	30	-45.2	-1,357.0	0
independent_fresh	primitive	0	11	30	0.0	0.0	0
independent_fresh	primitive	1	22	30	0.0	0.0	0
independent_fresh	primitive	2	33	30	0.0	0.0	0
independent_fresh	primitive	3	44	30	0.0	0.0	0
independent_fresh	primitive	4	55	30	0.0	0.0	0
independent_fresh	primitive	5	66	30	0.0	0.0	0

Workload	Condition	Stream	Seed	Boxes	Mean saving per box	Cumulative saving	Quality losses
recurring_family	frozen_exact	0	11	30	44.6	1,337.0	0
recurring_family	frozen_exact	1	22	30	39.9	1,198.0	0
recurring_family	frozen_exact	2	33	30	-16.8	-504.0	0
recurring_family	frozen_exact	3	44	30	74.7	2,242.0	0
recurring_family	frozen_exact	4	55	30	68.4	2,053.0	0
recurring_family	frozen_exact	5	66	30	15.0	450.0	0
recurring_family	growing_exact	0	11	30	44.4	1,332.0	0
recurring_family	growing_exact	1	22	30	40.2	1,206.0	0
recurring_family	growing_exact	2	33	30	2.2	66.0	0
recurring_family	growing_exact	3	44	30	46.8	1,404.0	0
recurring_family	growing_exact	4	55	30	39.3	1,180.0	0
recurring_family	growing_exact	5	66	30	14.3	430.0	0
recurring_family	primitive	0	11	30	0.0	0.0	0
recurring_family	primitive	1	22	30	0.0	0.0	0
recurring_family	primitive	2	33	30	0.0	0.0	0
recurring_family	primitive	3	44	30	0.0	0.0	0
recurring_family	primitive	4	55	30	0.0	0.0	0
recurring_family	primitive	5	66	30	0.0	0.0	0

Notes.

Table S2H. Expt-18 family box-level diagnostic opportunities

Box index	Task	Ideal stage	Ideal saving	Meets floor
0	inv_a8956340ad37	Not applicable	0.0	no
10	inv_41ec58701360	early	636.0	yes
20	inv_4dbbbf516584	early	132.0	yes

Box index	Task	Ideal stage	Ideal saving	Meets floor
30	inv_a21f4dc3bf54	early	36.0	no
40	inv_5ba647788a08	middle	132.0	yes
50	inv_6cf68140c244	early	84.0	yes

Notes.

Table S3. Cost definitions by experiment

Experiment	Deployment effort	Acquisition or learning cost	Diagnostic replay	Other recorded cost
Expt-4C	A on final-test boxes	Not in the held-out A	Not applicable	Internal execution is a separate column and is not added here
Expt-5	Primitive evaluations + retrieval + applicability + learned site checks + gate checks	Counterfactual gate labels, excluded from eval A	Not applicable	Not recorded as a second ledger in this table
Expt-7 / 7B	A, with consultation itemized	Ranker trained under the 1,600-cap labels	Recall rollouts are diagnostic	Budget 0 reproduces primitive A
Expt-8C	Not tested as a whole-box policy	Model fit is offline	Skip and commit remaining-solves	Feature checks: total in the ledger; rejected split is report-derived
Expt-9–14	A = primitive evaluations + lookup + validation + learning, as each file's A_identity records	Table construction is prior to fresh A except Expt-14 insertion, which is inside growing A	Not applicable	Storage bytes are in offline summaries where saved
Expt-15–16	Agent A on evaluation boxes	Region search, detection, validation, insertion	Expt-16 counterfactuals	Detector cell visits are not primitive evaluations
Expt-17	Not a deployed memory policy	Group checks during extraction, not placement evaluations	Not applicable	Matching checks are a separate census ledger
Expt-18	Not tested	Not re-extracted	Both branches of every sampled placement	Discovery checks are separate from counterfactual A
Expt-19	Not tested; commitment disabled	Dictionary is the frozen Expt-17 artifact	Reference counterfactuals, excluded from recognition cost	Recognition bills inventory, anchors, geometry, cache, and lookup

Notes.

Table S4. Diagnostic controls, audits, and interpretation limits

Control or limit	What was checked	Result	Source class

Control or limit	What was checked	Result	Source class
Expt-4C lesion	Removing the library	Matched the primitive trace	computed in the confirm run; see Table 2A note
Expt-4C transfer	A library used across seeds	Matched the hierarchical condition on the saved means	report-derived from the confirm report's identical published means
Expt-7B commitments	Whether the frozen ranker placed a library action	0 committed decisions in 767 decision rows across 48 seed-task pairs	computed
Expt-7 operation versus action recall	Whether a shortlisted operation is the same thing as a placement	Operation recall is not placement recall	design
Expt-8 label budget	Whether pair_eval_cap was the remaining-solve budget	Expt-8 already used eval_cap minus prefix effort. Expt-7 labels used the 1,600 cap. Expt-8B replay changed 0 fit rows and 0 validation rows	report-derived from EXPERIMENT_HISTORY.md
Expt-8B holdout	Whether eight boxes were unseen confirmation	They are development boxes. Expt-8C is the unseen set	design
Expt-9–14 first write	Whether seeds keep separate best solutions	One shared table; first write wins; overwrites are counted and not stored	design
Expt-11 sample size	What 33 rows are	11 tasks from four families × 3 seeds	design
Expt-16 fillings	Whether the primitive found the five fillings	It missed those five. It did not miss none. Most regions were impossible	computed
Expt-18 oracle	Whether box savings add candidate suffixes	No. The box figure is one selected candidate	computed
Expt-19 anchors	Anchor misses inside the sampled reference	None in that sample. Coverage of all useful placements was not measured	computed
Expt-19 gate	Whether recall can waive the cost gate	No. The recognizer failed 0.10. Commitment was not tested	computed

Notes.